

RDCC–Euclid Interface Paper V2.0: A Survey-Ready Framework for Testing Relaxation-Driven Cosmology

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Abstract

The Relaxation-Driven Cyclic Cosmology (RDCC) modifies the gravitational potential through a scale- and redshift-dependent relaxation kernel $\mathcal{R}(k, z; \alpha)$. These modifications induce geometric, morphological, and topological signatures in weak-lensing observables that are directly accessible to Euclid’s survey pipeline. This paper provides a survey-ready, Euclid-focused overview of RDCC. We present: (i) the theoretical structure of the relaxation kernel, (ii) the mapping from RDCC modifications to Euclid observables, (iii) Euclid-level forecasts for the relaxation parameter α , (iv) the degeneracy structure and multi-probe synergies, and (v) a practical integration pathway for incorporating RDCC into Euclid’s forward modeling, summary-statistics, and likelihood frameworks. The goal of this document is to provide a concise, technically precise reference for Euclid collaborators and theorists working on survey-level tests of relaxation-driven cosmology.

1 Introduction

The Euclid mission enables high-precision tests of gravitational physics through weak-lensing and galaxy-clustering measurements across a wide range of angular scales and redshifts. The Relaxation-Driven Cyclic Cosmology (RDCC) provides a minimal extension of Λ CDM in which the gravitational potential responds to large-scale structure through a relaxation mechanism governed by a single parameter α . This mechanism introduces scale-dependent deviations in the lensing potential that propagate into geometric and topological features of the convergence field.

Euclid’s tomographic binning, angular resolution, and survey depth make it particularly sensitive to RDCC-induced distortions. These distortions appear in power spectra, peak and void statistics, Minkowski functionals, persistent homology, and transport-based topology. The combination of these probes allows Euclid to constrain α at the percent level.

This paper provides a survey-ready interface between RDCC theory and Euclid analysis pipelines. Section 2 summarizes the theoretical structure of RDCC. Section 3 describes RDCC signatures in Euclid observables. Section 4 presents Euclid-level forecasts. Section 5 analyzes degeneracies and multi-probe synergies. Section 6 outlines integration pathways into Euclid pipelines. Section 7 provides a concise conclusion.

Nomenclature

α	RDCC relaxation parameter; controls scale-dependent modifications of the potential
Φ	Gravitational potential
$\Phi_{\Lambda\text{CDM}}$	Standard potential in the ΛCDM model
Φ_{RDCC}	Modified potential in the RDCC model
$\mathcal{R}(k, z; \alpha)$	RDCC relaxation kernel; scale- and redshift-dependent modification
k	Wavenumber in Fourier space
z	Redshift
κ	Weak-lensing convergence field
γ	Shear field
$P_\kappa(\ell)$	Convergence power spectrum as a function of multipole ℓ
MF_i	Minkowski functionals (area, boundary length, Euler characteristic)
$\mathcal{D}$	Persistence diagram in persistent homology
W_2	Quadratic optimal-transport distance between persistence diagrams
$\sigma(\alpha)$	Forecasted uncertainty on the RDCC parameter α
$\Delta \log Z$	Evidence contrast between RDCC and ΛCDM
v_k	Fisher channel (eigenvector of the information-tensor decomposition)
λ_k	Eigenvalue of the k -th Fisher channel (information contribution)

2 RDCC: Theoretical Overview

RDCC modifies the gravitational potential through a relaxation kernel

$$\Phi_{\text{RDCC}}(k, z) = \Phi_{\Lambda\text{CDM}}(k, z) \mathcal{R}(k, z; \alpha), \quad (1)$$

where $\mathcal{R}$ introduces scale-dependent deviations that become relevant at $k \gtrsim 0.1 h \text{ Mpc}^{-1}$. These deviations propagate into the lensing potential and affect a wide range of weak-lensing observables, including the convergence field κ , shear correlations, peak and void statistics, Minkowski functionals, persistent homology, and transport-based topology.

RDCC is fully compatible with Euclid’s tomographic binning and angular resolution. Its signatures are strongest at intermediate multipoles accessible to Euclid’s VIS instrument.

3 RDCC Signatures in Euclid Observables

RDCC induces characteristic distortions in Euclid weak-lensing data. These distortions arise from the scale-dependent modification of the lensing potential and propagate into geometric, morphological, and topological observables.

3.1 Power Spectra

The convergence power spectrum $P_\kappa(\ell)$ exhibits moderate-scale suppression or enhancement depending on the value of α . These effects are partially degenerate with ΛCDM parameters but remain distinguishable through tomographic and multi-probe combinations.

3.2 Peak and Void Statistics

RDCC shifts peak heights and void depths in the convergence field. These statistics provide strong sensitivity to relaxation-driven potential changes and are robust against shape noise at

Euclid depth.

3.3 Minkowski Functionals

The Minkowski functionals MF_i capture morphological distortions in area, boundary length, and Euler characteristic. RDCC induces characteristic departures from the Gaussian predictions of Λ CDM.

3.4 Persistent Homology

RDCC modifies the birth–death structure of topological features in the convergence field. Persistent homology provides a noise-resistant probe of these modifications and is highly sensitive to the relaxation kernel.

3.5 Transport-Based Topology

Optimal-transport distances W_2 between persistence diagrams provide a high-sensitivity probe of RDCC-induced non-Gaussian structure. These distances capture global topological rearrangements that are difficult to mimic with standard cosmological parameters.

4 Euclid-Level Forecasts

Using Euclid-like noise, resolution, and tomographic binning, RDCC forecasts yield

$$\sigma(\alpha) \sim \mathcal{O}(10^{-2}), \quad (2)$$

with the strongest constraints arising from:

- transport-based topological statistics,
- peak and void counts,
- neural-compressed summaries,
- combined multi-probe likelihoods.

Fisher-channel analysis shows that approximately 70% of RDCC information resides in the leading channel. This enables efficient compression and robust inference even in high-dimensional summary-statistics spaces.

5 Degeneracy Structure and Euclid Synergy

RDCC variations can partially mimic Λ CDM parameter shifts. Euclid’s multi-probe design helps break these degeneracies:

- **Tomographic binning** separates redshift-dependent relaxation effects.
- **Cross-correlation with galaxy clustering** reduces null directions.
- **Topological probes** provide strong orthogonality to Λ CDM-aligned manifolds.

The combination of geometric, morphological, and topological probes yields a highly complementary constraint structure.

6 Integration into Euclid Pipelines

RDCC can be tested within Euclid’s existing analysis framework through modular extensions at the forward-modeling, summary-statistics, and likelihood levels.

- **Forward modeling:** RDCC can be incorporated into lensing simulations via modified potential kernels $\Phi_{\text{RDCC}}(k, z)$.
- **Summary statistics:** RDCC signatures are compatible with Euclid’s power-spectrum pipeline and can be extended to include peaks, Minkowski functionals, persistent homology, and transport-based topology.
- **Likelihood integration:** RDCC likelihood components can be added modularly to Euclid’s cosmological parameter estimation framework.
- **Validation:** The RDCC Benchmark Suite provides simulation-based calibration and cross-probe consistency tests.

7 Conclusion

RDCC provides a theoretically motivated and observationally testable extension of Λ CDM. Its geometric and topological signatures are well matched to Euclid’s weak-lensing capabilities, and Euclid-level forecasts demonstrate that RDCC can be constrained with percent-level precision. This paper outlines a practical pathway for integrating RDCC into Euclid’s analysis pipelines and provides a survey-ready framework for testing relaxation-driven cosmology.